Exact Causal Simulation and the Limits of Informational Inference

Jan Mikulík 13 November 2026

Abstract. This synthesis connects four elementary distinctions: a physical state versus its record, a conditional law versus its realization, a possible action versus a realised outcome, and hidden information versus a gravitational horizon. Its technical core is a finite causal-seed theorem, adaptive stability, and a sharp attainable 2/3 observation-time exponent. Nested models, agency, and temporal receivers fit this framework when their physical assumptions are explicit. They do not independently derive those assumptions. Entropies are in bits; ln is natural.

1. One law, many possible questions

A finite controlled tree has complete histories h, actions a, responses y, and kernels $K(y \mid h, a)$. A deterministic policy chooses $a = \pi(h)$; a total response strategy chooses an output at every address (h, a). A common seed S and decoder $D(S, h, a)$ are exact when, for every policy, the transcript law is

$$P_\pi(h_n) = \prod_{t=0}^{n-1} K(y_{t+1} \mid h_t, \pi(h_t)).$$

Existence is simple: sample independent cells $S_{h,a} \sim K(\cdot \mid h, a)$ and read the current cell. Reaching h depends only on earlier cells, so conditioning on that event preserves the current law. The same address reads the same answer for every fixed seed, regardless of the policy. This table can be much more expensive than necessary.

For the finite-valued transcript information $I_\pi = -\log_2 P_\pi(Z_\pi)$, put

$$F_*(u) = \min_\pi \mathbb{P}(I_\pi \le u), \qquad L = \int_0^\infty (1 - F_*(u))\, du.$$

There are finitely many policies, so F_* is a CDF. Let Z_* have this law. If C is the minimum seed entropy, the causal theorem [1] is

$$\boxed{L \le C \le L + c_*, \qquad c_* = (1 + \log_2 e)/2 \approx 1.22135.}$$

The numerical constant is already known from minimum-entropy coupling [2]; the causal requirement is agreement at shared histories. No claim that this is the smallest possible universal constant is needed here.

Proof. A seed atom of mass w maps to a transcript atom of mass at least w. Thus seed information $J = -\log_2 w$ dominates every I_π, giving $F_J \le F_*$ and $H(S) \ge L$. For the upper bound, let $p(v)$ be the path mass and $\chi_d(v)$ indicate agreement with strategy d. Exactness means $\sum_j w_j \chi_{d_j}(v) = p(v)$ for every history. Both p and χ_d obey $\sum_y r(vay) = r(v)$ for every action.

For a nonnegative residual flow r, set $B_r(v) = r(v)$ at leaves and $B_r(v) = \min_a \max_y B_r(vay)$ elsewhere. Induction on the tree shows that $B_r(\emptyset)$ is both the largest extractable strategy weight and the minimum, over policies, of their largest residual leaf mass. Extract this weight and subtract its strategy flow. Some positive leaf vanishes, so extraction terminates; weights decrease and sum to one.

After extracting every weight greater than δ, some policy has all residual leaves at most δ. A leaf of original mass $p > \delta$ must already have lost a weight greater than δ. Its residual is at most $\min(\delta, p - \delta)$; for $p \le \delta$ it is at most p. Define the nondecreasing function $\psi(u) = u$ for $u \le 1/2$, $\psi(u) = 1/2$ for $1/2 < u < 1$, and $\psi(u) = 1$ for $u \ge 1$. These residual bounds are at most $p\psi(\delta/p)$. Since I_π is stochastically dominated by Z_*, the greedy seed satisfies

$$\mathbb{P}(J \ge t) \le \mathbb{E}\psi(2^{Z_* - t}), \qquad \int_0^\infty \psi(2^{z-t})\, dt = z + \frac{1}{2} + \frac{1}{2 \ln 2} \quad (z \ge 0).$$

Integrating proves $H(S) \le L + c_*$. A minimum exists by compactness of the exact laws on the finite set of total strategies. Zero-probability branches can be pruned and the decoder extended arbitrarily there. □

2. What can be distinguished, and how fast?

Adaptive equivalence. Two models on a common finite action-response tree have identical transcript laws under every policy exactly when their next-response kernels agree at every history reachable in either model. The forward implication follows by recovering prefix probabilities from terminal laws and taking

$$K(y \mid h, a) = \mathbb{P}(hay)/\mathbb{P}(h) \qquad (\mathbb{P}(h) > 0).$$

The reverse implication is multiplication along each path. This is the chain rule on complete histories; it requires no extra finite-memory Markov assumption. Kernels on histories unreachable in both models cannot be identified.

Stability. If, at step t, the kernels differ by at most ε_t in total variation at every common reachable history and action, then

$$\sup_B |P_\pi(B) - Q_\pi(B)| \le 1 - \prod_{t=1}^{n}(1-\varepsilon_t) \le \min\left\{1, \sum_{t=1}^{n}\varepsilon_t\right\}.$$

Couple equal replies with mass $\sum_y \min(p_y, q_y) = 1 - \mathrm{TV}(p, q)$. Until replies separate, the policy selects the same action; the probability of agreement throughout is at least the displayed product. The sum is a union-bound relaxation. For $n = 10$, $\varepsilon_t = 0.01$, the product bound is 0.09562.

Why marginals do not suffice. Let B be fair. The records (B, B) and $(B, 1-B)$ have identical one-time marginals but disjoint supports, hence joint TV distance one. Their second-response kernels are opposite. Equal adaptive kernels, by contrast, force equal joint laws. TV itself never exceeds one; only an untruncated upper estimate $\sum_t \varepsilon_t$ can do so. The sum relaxation is first-order accurate for small total error, but small error is not a condition for validity or the only case of equality.

A quantitative revelation model. Let P be an orthogonal projection in $\mathbb{R}^d$, $Q = I - P$, B orthogonal, and $\|PBQ\| \le \varepsilon$. Take a finite hidden input $QW = W$ with variance $V = \mathbb{E}\|W - \mathbb{E}W\|^2 \le 1$, and passive observations

$$Y_t = PB^t W + \sigma N_t,$$

with independent standard Gaussian N_t, independent of W, fixed $\sigma > 0$, and fixed sampling interval. Then

$$I(W; Y_{1:n}) \le \min\left\{H(W), \frac{V}{2\sigma^2 \ln 2}\sum_{t=1}^{n}\min(1, t^2\varepsilon^2)\right\} \le \frac{\varepsilon^2 n(n+1)(2n+1)}{12\sigma^2 \ln 2}.$$

Indeed, $C_t = PB^tQ$ obeys $C_{t+1} = PBPC_t + PBQQB^tQ$, so $\|C_t\| \le \min(1, t\varepsilon)$. For a Gaussian channel $Y = AW + \sigma N$, comparison with the Gaussian of mean $A\mathbb{E}W$ gives

$$I(W; Y) \le \sum_w p_w D_2(N(Aw, \sigma^2 I) \,\|\, N(A\mathbb{E}W, \sigma^2 I)) = \frac{\mathbb{E}\|A(W - \mathbb{E}W)\|^2}{2\sigma^2 \ln 2}.$$

The inequality follows from splitting the average relative entropy into $I(W; Y)$ plus a nonnegative relative entropy. Stack the n observation blocks and use $\sum t^2 = n(n+1)(2n+1)/6$ to obtain the claim.

Sharp attainable time scale. For fixed $0 < \eta < 1$ bit, let n_* be the smallest n for which some such model reveals at least η. As $\varepsilon \downarrow 0$ with σ fixed,

$$\boxed{n_* = \Theta((\sigma/\varepsilon)^{2/3}).}$$

The information bound gives $n \ge (2\eta \ln 2)^{1/3}(\sigma/\varepsilon)^{2/3}$. To attain this order, choose a planar rotation through $\theta = \arcsin\varepsilon$, project onto the first coordinate, and set $W = (0, V_0)$ with $V_0 = \pm 1$ fair. Then $Y_t = -V_0 \sin(t\theta) + \sigma N_t$. Matched filtering gives $T = s_n V_0 + G$, where

$$s_n^2 = \sigma^{-2}\sum_{t=1}^{n}\sin^2(t\theta), \quad G \sim N(0,1), \qquad I(W; Y_{1:n}) \ge 1 - h_2(\Phi(-s_n)).$$

The last bound follows by thresholding T and bounding conditional binary entropy by the entropy of its error indicator. For $n = \lceil C(\sigma/\varepsilon)^{2/3} \rceil$, $n\theta \to 0$ and $s_n^2 \to C^3/3$. Large fixed C reaches any fixed $\eta < 1$. The exponent is $2/3$ because amplitude grows like $t\varepsilon$ and accumulated discrimination energy like $\varepsilon^2 n^3$, not $\varepsilon^2 n^2$. This is best attainable behavior in the class, not a guarantee for every receiver. Without noise, the first rotational readout reveals the bit exactly. To apply the finite-tree theorem to these continuous observations, first record finitely many output bins.

3. A present record, imagined futures, and effective choice

Observation is a quotient, not automatically destruction. A property $f(x)$ is recoverable from $O(x)$ exactly when it is constant on each fibre of O. For a finite classical random vector and complementary projections,

$$Z = PZ + QZ, \qquad H(Z) = H(PZ) + H(QZ \mid PZ).$$

This follows because $Z \leftrightarrow (PZ, QZ)$ is invertible. A constant readout can leave hidden distinctions: if X is a fair choice between two states and $O(X) \equiv 1$, then $H(O(X)) = 0$ but $H(X \mid O(X)) = 1$. Zero entropy means certainty, not zero value or nonexistence. The map $x \mapsto (x, O(x))$ retains the source. A quantum instrument, not an abstract classical projection, determines actual measurement disturbance.

Time after time. Let τ denote physical time, c an internal clock readout, and k a model-refinement index. At a fixed data snapshot $R = r$, the observer may iterate $m_{k+1} = \mathcal{A}_r(m_k)$. Physical implementation takes time and energy; k is not thereby a second physical temporal dimension. If W is an unknown target and refinements use only R and randomness U independent of (W, R), then

$$M_k = f_k(R, U) \quad \Longrightarrow \quad I(W; M_k \mid R) = 0, \qquad I(W; M_k) \le I(W; R).$$

Conditional independence and the information chain rule prove both claims. Computation can expose implications already present in the record; new evidence requires a new channel.

A concrete example is a fair source labelled either by S or by $T = 1 - S$, with each representation decoding to the same fair output. Every observation has the same likelihood under both descriptions, so their posterior odds remain their prior odds. Refinement can change representation without selecting a uniquely correct label. A contraction can converge even to an incorrect fixed point. Convergence needs a separate correctness argument.

Relational time is a correlation. For orthogonal clock states, unitary U with $U^N = I$, and normalized ψ_0, the history state [3]

$$|\Psi\rangle = N^{-1/2} \sum_{t=0}^{N-1} |t\rangle \otimes U^t |\psi_0\rangle$$

is normalized and invariant under $T_C \otimes U$. Conditioning on clock value t gives $U^t\psi_0$ with clock probability $1/N$. Orthogonality proves normalization; reindexing proves invariance. A physical calibration, such as $U = e^{-iH\Delta\tau/\hbar}$, is additional input. Adding memory or an instrument changes the dynamics and need not preserve cyclicity.

Imagination and agency. At history h, candidate policies determine possible future laws $P_\pi(\cdot \mid h)$. Their supports are the model's possible continuations. Imagining one does not make it a physically actual timeline. An independent fair probe A selecting an address on a fixed answer sheet S gives

$$H(A \mid S) = 1, \qquad H(Y \mid S, A) = 0.$$

This formalizes an open choice relative to that sheet, the stipulated meaning of "fundamental chaos." It does not prove ontological randomness or free will. The seed theorem itself shows that finite stochastic laws admit deterministic responses conditional on a random seed.

Operational agency can require represented alternatives, a policy responsive to reasons, and causal efficacy: different interventions change future laws. Randomness alone does not provide these. If A is fair but Y is independent of every intervention on A, there is randomness without efficacy. If $Y = A$ but the current policy always chooses zero, its action entropy is zero although intervening to choose one changes the outcome. A metaphysical account of freedom requires further premises beyond this behavioral contract.

A temporal receiver needs a physical channel. A causal response

$$y(\tau) = \int_0^\infty K(u) s(\tau - u)\, du + \eta(\tau)$$

represents a present trace of previous inputs. A microtubule implementation must specify s, its coupling, response, noise, and an independently predicted downstream effect. This is a proposed empirical test, not a consequence of the entropy theorem. Resonance alone does not distinguish quantum reception from a classical response. "All-knowing now" can consistently mean complete conditional predictions for a specified experiment family; it need not mean deterministic outcomes or knowledge of all reality.

4. Information access is not gravitational geometry

Encoding does not select a physics. For an injective state encoding E and physical update F, the equivalent update is $\widetilde{F} = E \circ F \circ E^{-1}$ on the image. This changes coordinates, not the physical law. Scalar entropy is even less specific: the state $\rho = I/2$ has one bit for every Hamiltonian $H_\lambda = \lambda|1\rangle\langle 1|$, while $\mathrm{tr}(\rho H_\lambda) = \lambda/2$. Information quantity alone does not specify energy, spatial distribution, or gravity.

Three distinct artificial horizons.

Access horizon	A hidden bit is retained internally; every allowed external query returns a fixed marker. Every external transcript is independent of the bit. Adding a read-cell intervention removes this protocol-relative restriction.
Analogue horizon	A specified wave mode cannot propagate out through a boundary in a medium. Sonic-horizon theory and experiments provide examples [4,5]. This need not trap light or all excitations.
Gravitational horizon	In an asymptotically flat spacetime, the boundary of $\mathcal{M} \setminus J^-(\mathcal{I}^+)$ separates events unable to signal to future null infinity. Formation requires a physical spacetime geometry, not merely inaccessible data.

A reversible transfer $(b, 0) \mapsto (0, b)$, extended to SWAP, makes one register constant while preserving the bit in the other. This constructs the first kind of concealment without global erasure. It neither manufactures a gravitational hole nor resolves the black-hole information problem.

For Schwarzschild geometry, $r_s = 2GM/c^2$; gravitational thermodynamics gives [6,7]

$$S_{\mathrm{BH}} = \frac{k_B A_{\mathcal{H}}}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}.$$

These concern a physically specified horizon. Assigning entropy to a file supplies no formation mechanism. An actual gravity proposal must supply stress-energy, field equations, and formation dynamics. The finite-seed theorem does not select the Born rule, a Hamiltonian, Einstein's equations, or the Standard Model.

The logical boundary. Equality of images, $O(x) = O(x')$, is compatible with $x \neq x'$. Literal $0 = 1$ in a unital ring instead forces $a = a1 = a0 = 0$ for every a: a trivial algebra, not ordinary arithmetic with distinct numbers. Four-valued evidence may designate both a proposition and its negation without explosion, but that changes the semantics; it is not ordinary equality with unrestricted substitution. None of these statements implies that understanding destroys a person, that hidden entropy constitutes life, or that affection is a unique mathematical operator. Respect for unobserved experience and consent can be an explicit philosophical commitment; it is not an entropy theorem.

Scope and efficiency. The common-seed construction resembles common random numbers, but requires consistency at every shared history. A policy intervention here is specified by the controlled kernels; it is not causal identification from observational data. Shannon seed entropy is an ensemble cost, not Kolmogorov complexity of an individual description. Algorithmic efficiency would require a stated input representation and polynomial bounds on preprocessing, storage, and response time, including a model for exact real probabilities. None follows from a small entropy overhead. A compactly described depth-n experiment can have exponentially many histories.

Principle. A record determines an observational class; a policy determines which question is asked; a physical law determines the response. A complete description, an accessible description, and an effective intervention are different objects. These distinctions connect the whole framework without identifying possibilities with events or information barriers with gravitational horizons.

Provenance and references. This is a condensed synthesis, not a claim of a new universal entropy constant or a new physical theory. Proofs use ordinary mathematics; no fresh full Lean build is claimed. The weak-coupling calculation makes no priority claim. Prepared with assistance from OpenAI ChatGPT.

1. J. Mikulík, *One Seed for Every Adaptive Policy*, manuscript and Lean project (2026), `https://github.com/jendamikulik/finite-Shannon-theorem`; inspected revision `1ce5a1a78a4486431a08d949bf21df025000777f`.
2. S. Compton et al., Minimum-Entropy Coupling Approximation Guarantees Beyond the Majorization Barrier, PMLR **206**, 10445--10469 (2023).
3. D. N. Page and W. K. Wootters, Phys. Rev. D **27**, 2885--2892 (1983).
4. W. G. Unruh, Experimental Black-Hole Evaporation?, Phys. Rev. Lett. **46**, 1351--1353 (1981).
5. J. Steinhauer, Observation of quantum Hawking radiation and its entanglement in an analogue black hole, Nature Physics **12**, 959--965 (2016).
6. J. D. Bekenstein, Black Holes and Entropy, Phys. Rev. D **7**, 2333--2346 (1973).
7. S. W. Hawking, Particle creation by black holes, Commun. Math. Phys. **43**, 199--220 (1975).